Computational Complexity

Reading guide and Homework for **Weeks 10&11**

1 Reading Guide

This week we learned about cryptography and pseudorandomness:

- We learned about cryptographic schemes, the one-time pad, perfect security, and why cryptography is unavoidably tied to computational complexity. This is Section 9.1 of the book.
- We learned about one-way functions, pseudorandom generators, and their connection to cryptography. Section 9.2.
- We learned about Yao's theorem, that unpredictability implies pseudorandomness. The proof technique is often called the *hybrid method*. Section 9.3.
- We learned that one-way permutations imply the existence of PRGs, which is a toy version of the Goldreich–Levin theorem that one-way functions imply PRGs. Section 9.3.
- We learned about pseudorandom functions, weak derandomization, and bit commitment. Sections 9.5.1-9.5.3.

2 Homework

Skill: *Reason about cryptographic primitives.*

1. Do exercises 9.1–9.6 and 9.9–9.16 of the book.